



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Programs: Electrical Engineering

Course Title:	ELE632
Course Number:	Signal and Systems II
Semester/Year (e.g.F2016)	Winter/2026

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Assignment/Lab Number:	5
Assignment/Lab Title:	Sampling and the Discrete Fourier Transform

Submission Date	April 5, 2026
Due Date	April 5, 2026

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*By signing above you attest that you have contributed to this submission and confirm that all work you have contributed to this submission is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <https://www.torontomu.ca/content/dam/senate/policies/pol60.pdf>

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INTRODUCTION

This experiment will examine how to downsample several discrete signals, including an audio source, in this project and look at how the spectrum of the signal shifts. Furthermore, investigating that even if zero-padding rises, the frequency resolution doesn't produce fresh data. Therefore, given the quantity of samples in the time domain is insufficient, zero-padding will not adequately describe frequency components. Additionally, you will create an audio equalization and work with a signal in the frequency domain. Thus, examining the impact on the time-domain signal.

Previously, in Lab 1, recursive methods were used to determine the physical attributes (such as transformations, energy, and power) of discrete signals using Python algorithms and data structures. In lab 2, the basic concepts of discrete-time signal processing to analyze mathematical models and digital implementations to calculate the unit impulse response, zero-input response, zero-state response, and total response for linear time-invariant discrete-time (LTID) systems. Lab 3 allowed students to have an opportunity to learn about the Discrete-Time Fourier Series (DTFS) through the creation of a program in Python that implements the polyphase decomposition of the DTFS, as well as learning about the properties associated with the DTFS. As for Lab 4, the Discrete-Time Fourier transform (DTFT) will be covered. Students will discover how to compute a signal's DTFT using the FFT and look at the DTFT's time convolution property. Additionally, you will create a FIR high-pass filter and examine the distinction between a realizable filter and an ideal filter.

Throughout these experiments, the following student number will be computed as per the given sequence of variables.

$$\begin{aligned} \text{Student ID} &= 501103255 & [0] \\ [A = 5, B = 0, C = 1, D = 1, E = 0, F = 3, G = 2, H = 5, I = 5] \end{aligned}$$

Equation 0: Assigned Student Number Sequence

PREPERATION

The Discrete Fourier Transform (DFT) is defined as follows where both x_n and X_r are periodic sequences with period N_0 . If $x[n]$ is the n th sample from a signal $x(t)$, the DFT of xn is defined as below:

$$X_r = \sum_{n=0}^{N_0-1} x_n e^{-jr\Omega_0 n} \quad [1]$$

Equation 1: X_r is the r th sample from its Fourier transform $X(\omega)$

- Where $\Omega_0 = \omega_0 T = \frac{2\pi}{N_0}$

$$x_n = \frac{1}{N_0} \sum_{n=0}^{N_0-1} X_r e^{jr\Omega_0 n} \quad [2]$$

Equation 2: The Inverse Discrete Fourier Transform (IDFT)

When attempting to perform `Audio(y, rate=fs)`, a `Traceback` error appears. The `soundfile` library frequently reads audio in a format where samples are rows and channels are columns, but `IPython.display` does not. This is a technical "shape mismatch" problem. The `audio` function anticipates the contrary. The WAV file header includes a 16-bit field (the 'H' format) to contain the number of audio channels, which results in the particular error message, `error: 'H' format requires 0 <= number <= 65535`. The computer is misinterpreting your thousands of audio samples for the number of channels because your audio array is oriented wrongly. Since the number exceeds the 16-bit limit of 65,535, the code crashes while attempting to construct the audio header.

```
[ ] !sudo apt-get install libportaudio2
    !pip install sounddevice

[ ] import numpy as np
    import matplotlib.pyplot as plt
    from scipy import signal
    from scipy.fft import fft, ifft, fftshift
    import soundfile as sf

    from IPython.display import Audio

[ ] audio = '/content/chirp.wav' # make sure the audio file is in your working directory
    y, fs = sf.read(audio) # read in audio data (y) and sampling frequency (fs)

[ ] Audio(y, rate=fs) # play audio. be sure to specify the rate parameter as your sampling frequency
```

Image 0: ELE632_Lab_5_Playing_Audio_in_Colab.ipynb

The link between the sampling rate (f_s), which indicates samples per second, and the sampling period ($T = \frac{1}{f_s}$), which indicates the time interval between those samples, is essential to digital audio processing. Higher f_s is required to capture high-frequency information and prevent aliasing distortion, but it also increases data density and memory utilization. When using Google Colab to play `chirp.wav`, for example, the proper `rate= f_s` must be specified to guarantee that the audio plays at its natural pitch. However, as your error logs show, a shape mismatch can happen when the player misinterprets large sample counts as a huge number of audio channels. This issue can be fixed by flipping the data array so that it matches the software's expectations.

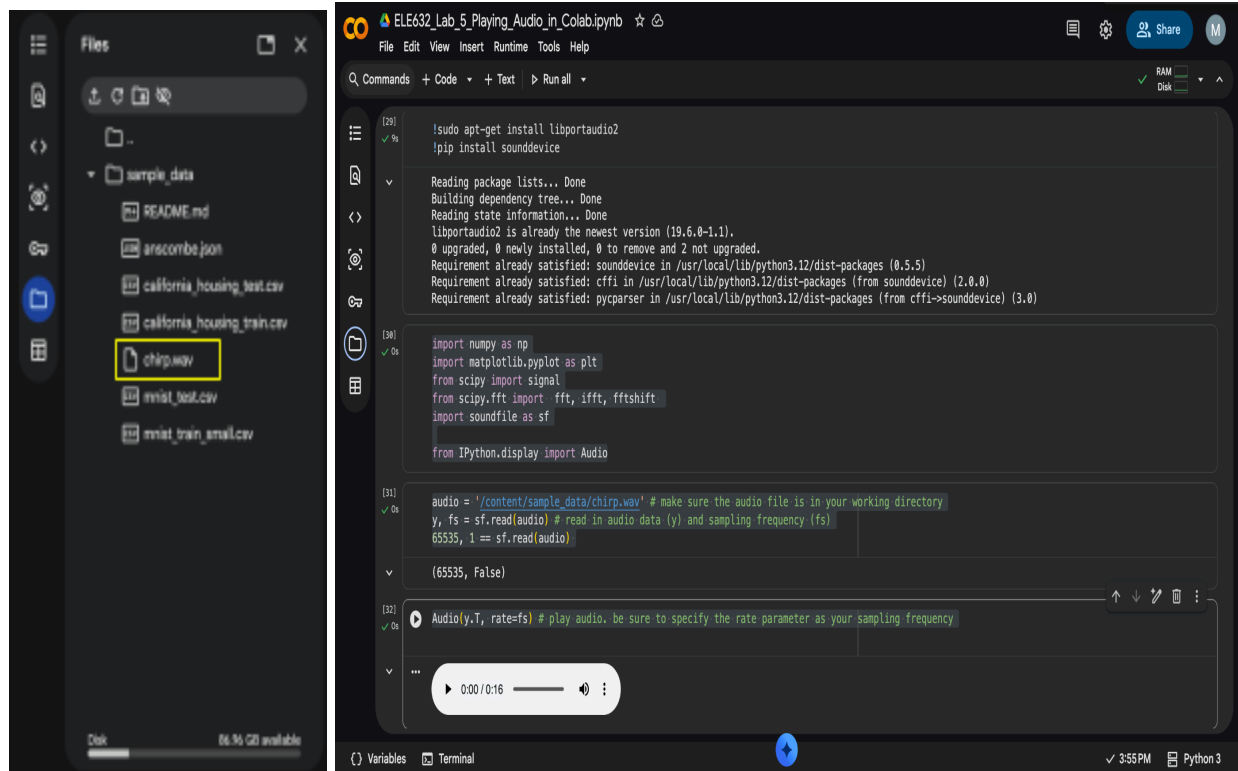


Image 1: Google Colab Workspace and Directory

A successful pip installation sequence on your terminal or notebook cell is seen in the second screenshot. It verifies that the required audio library for your lab was built up by running the command `!pip install sounddevice`. The output shows that the system downloaded and installed `sounddevice-0.5` after checking for requirements and determining that `cffi` and `pycparser` were already satisfied. This result, in contrast to the first screenshot, is error-free and only verifies that your environment now has the necessary tools to communicate with audio devices and process sound signals.

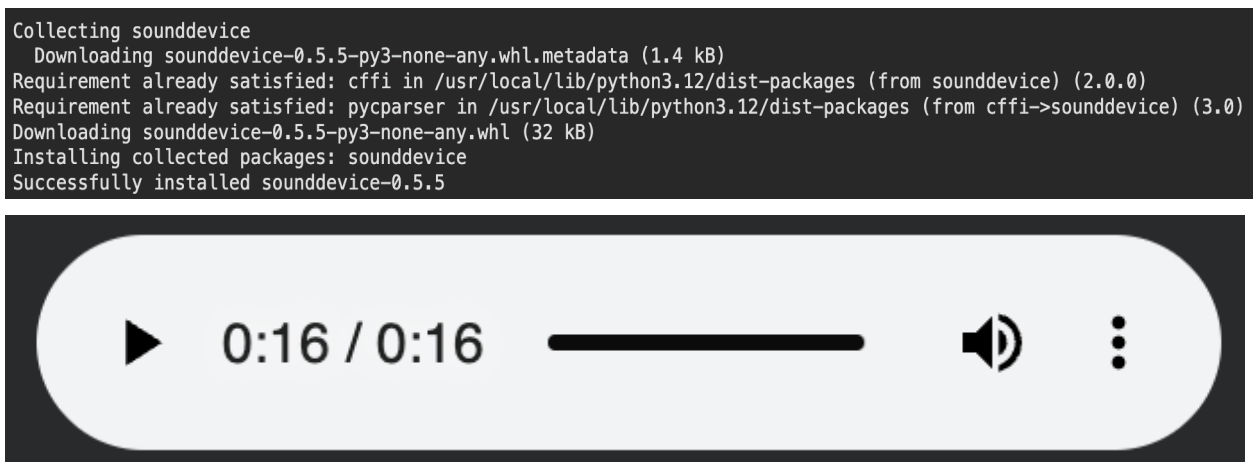


Image 2: Listening to Audio Files in Google Colab using Python3

PART A: Discrete Fourier Transform (DFT) & Zero Padding

Question 1:

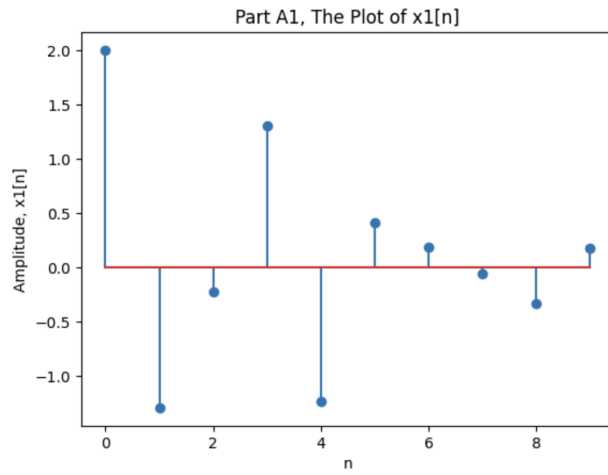


Figure 1: Plot of Signal $x_1[n]$

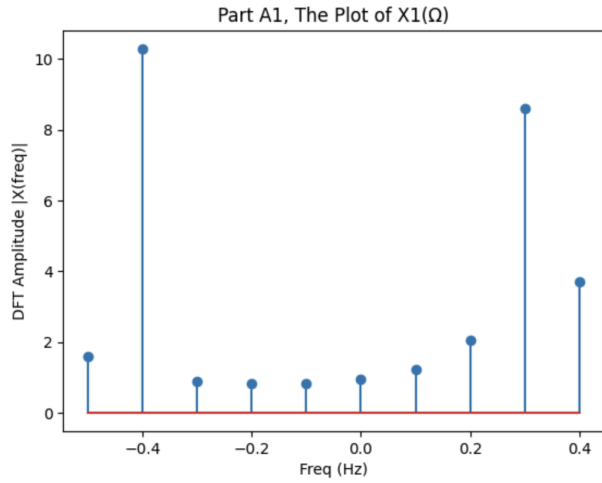


Figure 2: Magnitude Spectra ($X_1[\Omega]$)

$$x_1[n] = e^{j2\pi n \frac{(10(H+1))}{100}} + e^{j2\pi n \frac{33}{100}}$$

$$x_1[n] = e^{j2\pi n \frac{3}{5}} + e^{j2\pi n \frac{33}{100}}$$

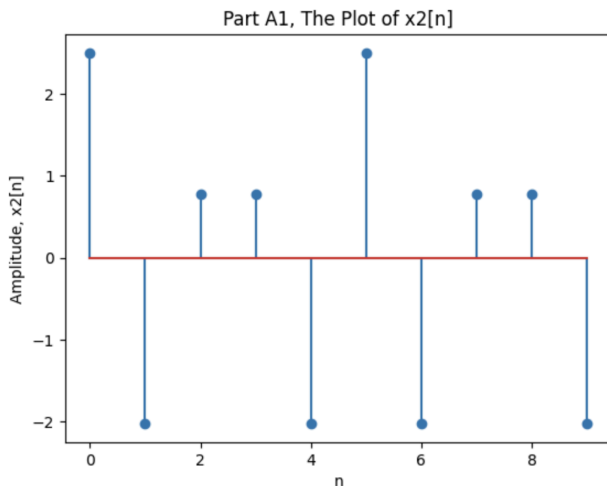


Figure 3: Plot of Signal $x_2[n]$

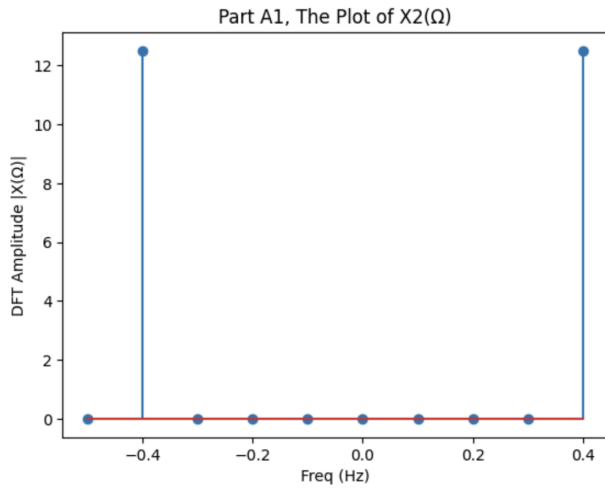


Figure 4: Magnitude Spectra ($X_2[\Omega]$)

$$x_2[n] = 2\cos(2\pi n \frac{(10(H+1))}{100}) + 0.5\cos(2\pi n \frac{(10(I+1))}{100})$$

$$x_2[n] = 2\cos(2\pi n \frac{60}{100}) + 0.5\cos(2\pi n \frac{60}{100})$$

$$x_2[n] = 2.5\cos(2\pi n \frac{3}{5})$$

From equation [0] above, the sequence of the assigned student ID suggests that the value for $H = 5$ and $I = 5$ as per the 9-digit number where I corresponds to the ninth letter of the alphabet.

$$x_1[n] = e^{j2\pi n \frac{3}{5}} + e^{j2\pi n \frac{33}{100}}$$

$$x_2[n] = 2.5 \cos(2\pi n \frac{3}{5})$$

I.

A signal which is symmetric across the y-axis in its Fourier Spectra is a symmetric signal. That is to say, x at each negative value n is the same for every positive counterpart. Signal x_1 does not accomplish this, while signal x_2 does. Therefore, only x_2 has a symmetric spectrum.

When analyzing the above spectrums, it appears that x_2 has a symmetrical signal while x_1 does not. This is due to the second function being based on the sum of two cos functions, which are both real signals, while the first signal is of two complex exponential terms. The complex exponential terms are not real, as shown in the magnitude spectra, which are not symmetrical, while cos is symmetrical and real.

II.

It is not possible to distinguish the frequency component of the first signal. Its peaks in its frequency plot are not symmetric, and thus the frequency at those peaks does not signify a repeating signal to begin with. This is not the case for the second signal, however, which may have frequencies present at those peaks.

It is possible to distinguish frequency components in x_2 , as they occur at a specific frequency. for x_2 , there should be an impulse at both positive and negative 4/10. On the other hand, the 2 impulses for x_1 seem to have been spread apart.

III.

This is because the signal is not periodic and cannot be expressed as a finite number of frequencies.

This is due to spectral leakage, where the values leak to other frequencies since the value at the complex exponential is not being sampled (sampling from $[-0.5 : 0.5 : 0.1]$ but complex exponentials exist at $\frac{33}{100}$ and $\frac{3}{5}$.

Question 2:

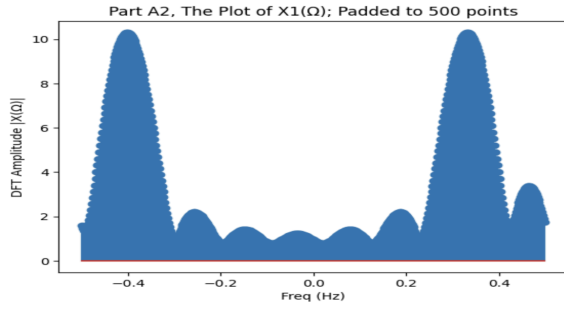


Figure 5: Magnitude Spectra ($X_1[\Omega]$)

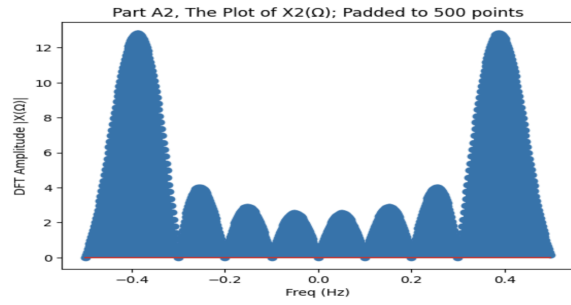


Figure 6: Magnitude Spectra ($X_2[\Omega]$)

The padded zeroes have the effect of spreading the frequency. This introduces frequencies which are not present in the original signal. This is not necessarily better and can even be undesirable (noise). Therefore, even after padding, the signals' spectra have not seemed to change.

Question 3:

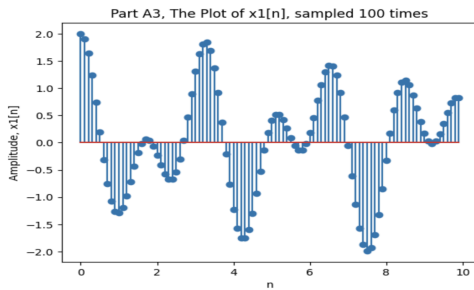


Figure 7: Plot of Signal $x_1[n]$

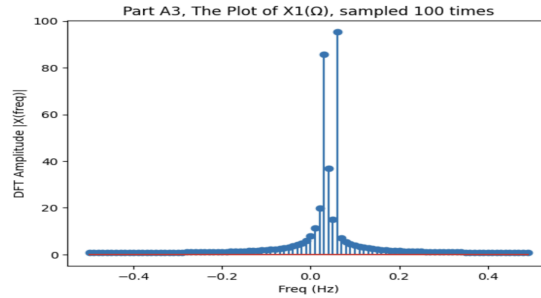


Figure 8: Magnitude Spectra ($X_1[\Omega]$)

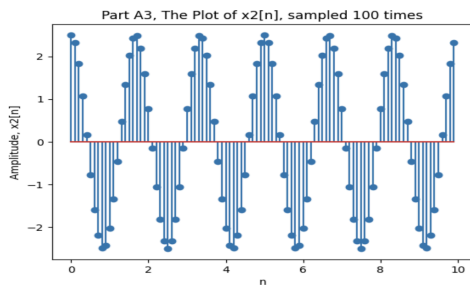


Figure 9: Plot of Signal $x_2[n]$

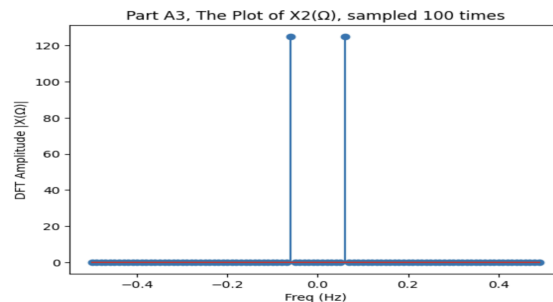


Figure 10: Magnitude Spectra ($X_2[\Omega]$)

By increasing the sample size, the spectra for $x_1[n]$ has resulted in only 2 impulses located at the frequencies $\frac{3}{5}$ and $\frac{33}{100}$ as expected. Therefore, by taking 100 samples has made it clear what the frequency components of the signal x_1 are. However, it is still clear that x_1 is not symmetric, while x_2 is.

Question 4:

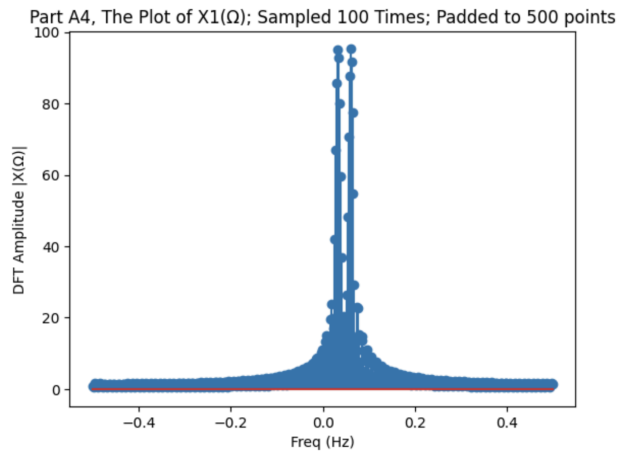


Figure 11: Magnitude Spectra ($X_1[\Omega]$)

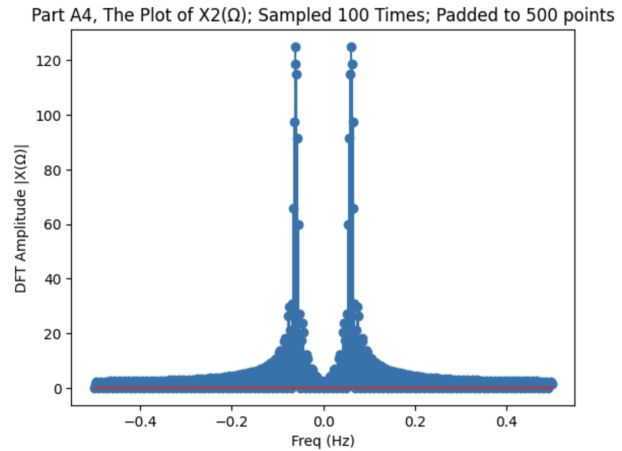


Figure 12: Magnitude Spectra ($X_2[\Omega]$)

While the frequencies are still spread out by padded zeros, and this is typically undesirable (adds noise), in this case, the increased sample rate causes the noise generated to be far less apparent. Thus, similar to before, there are no improvements to the spectra after adding padding to them (improving resolution)

PART B: Sampling

In this section, the sample of an audio signal in the time domain will be investigated, and how different sampling rates affect the frequency spectrum.

```
filename = 'audio.wav'
```

Image 3: Save the Audio File in the Project Directory Path

```
y, fs = soundfile.read(filename)
```

Image 4: Python *sounddevice* library to Play the audio file generated from the code, and *read* in *soundfile* reads .wav audio files and returns sampled data y and sampling frequency f_s

Question 1:

In order to get the number of samples, N_0 , just taking the length of y ($len(y)$) will provide the number of samples. For the duration of the signal T_0 , take the number of samples (N_0) and divide by f_s . Similarly, to get the Sampling Interval (Period), T , just take the reciprocal of the Sampling Frequency f_s .

$$N_0 = len(y) = 730148$$

$$T_0 = \frac{N_0}{f_s} = 16.56 \text{ secs}$$

$$T = \frac{1}{f_s} = 1.5625 * 10^{-5} \text{ secs}$$

```
Samples (N0): 730148
Length (T0): 16.556643990929704
Sampling Interval (T): 1.5625e-05
```

Figure 13: Number of Samples (N_0), Duration of the Signal (T_0), and Sampling Interval (T)

Question 2:

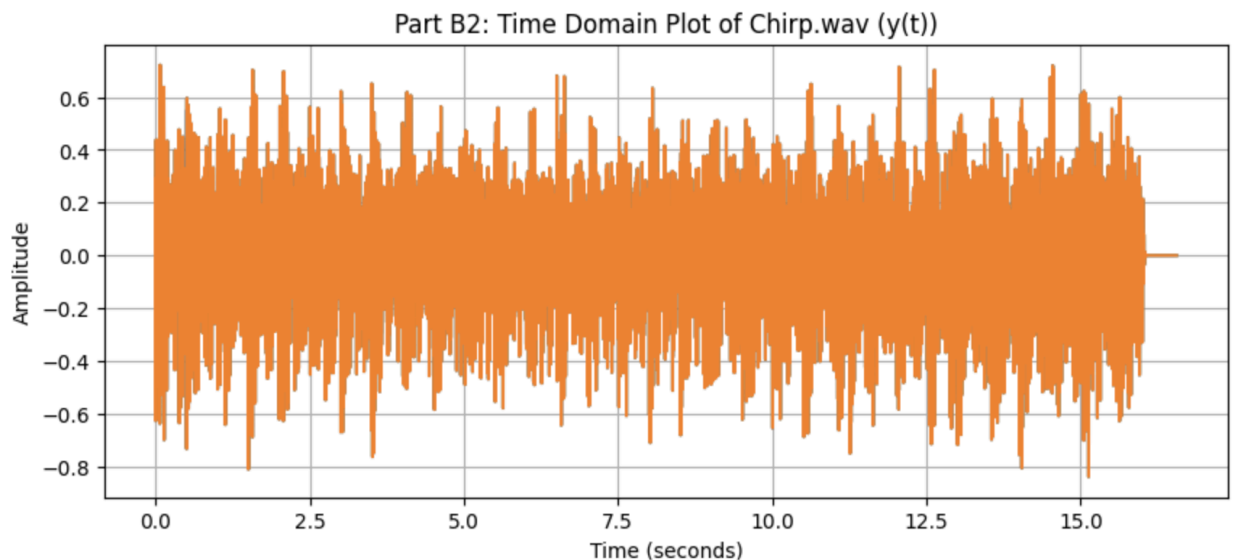


Figure 14: Time Domain Plot of chirp.wav ($y(t)$)

Question 3:

Part B3, The Plot of $Y(\Omega)$; The Frequency Spectrum of Chirp.wav

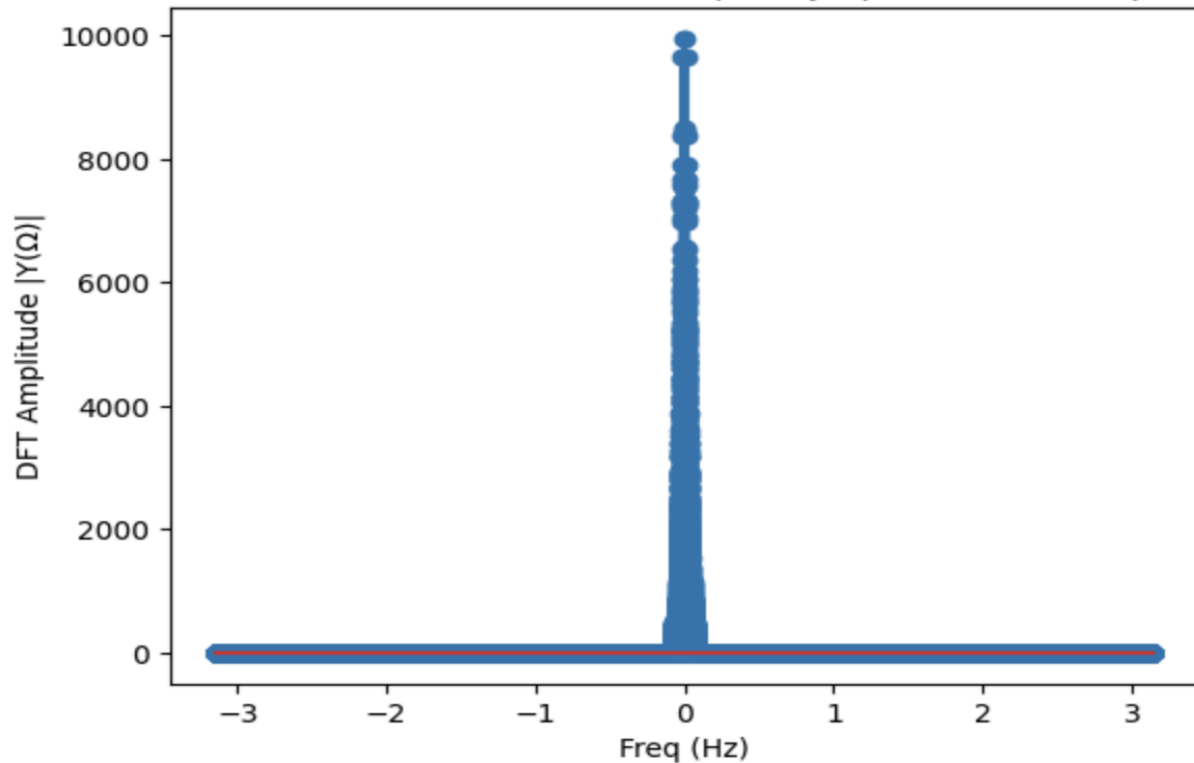


Figure 15: Discrete Fourier Transform (DFT) of $y[n]$ in the Frequency-Domain ($Y[\Omega]$)

Question 4:

Samples: 730148
Length: 16.556643990929704

Figure 16: Signal Parameters for $y_1(t)$

In order to get the number of samples, N_0 , just taking the length of y ($\text{len}(y)$) will provide us the number of samples. for the duration of the signal T_0 , Take the number of samples (N_0) and divide by f_s . Similarly, to get the Sampling Period T , just take the reciprocal of the Sampling Frequency f_s .

$$N_0 = \text{len}(y1) = 730148$$

$$T_0 = \frac{N_0}{f_s} = 16.56 \text{ secs}$$

$$T = \frac{1}{f_s/2} = 0.78125 * 10^{-5} \text{ secs}$$

Question 5:

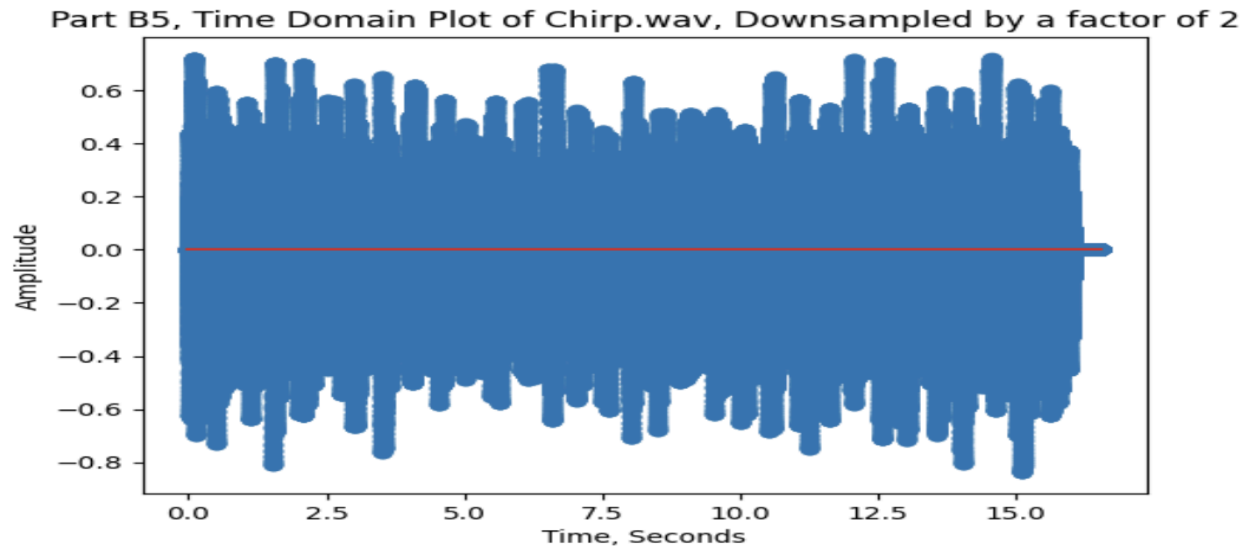


Figure 17: Time Domain Plot of chirp.wav ($y(t)$) Downsampled by a Factor of 2

Question 6:

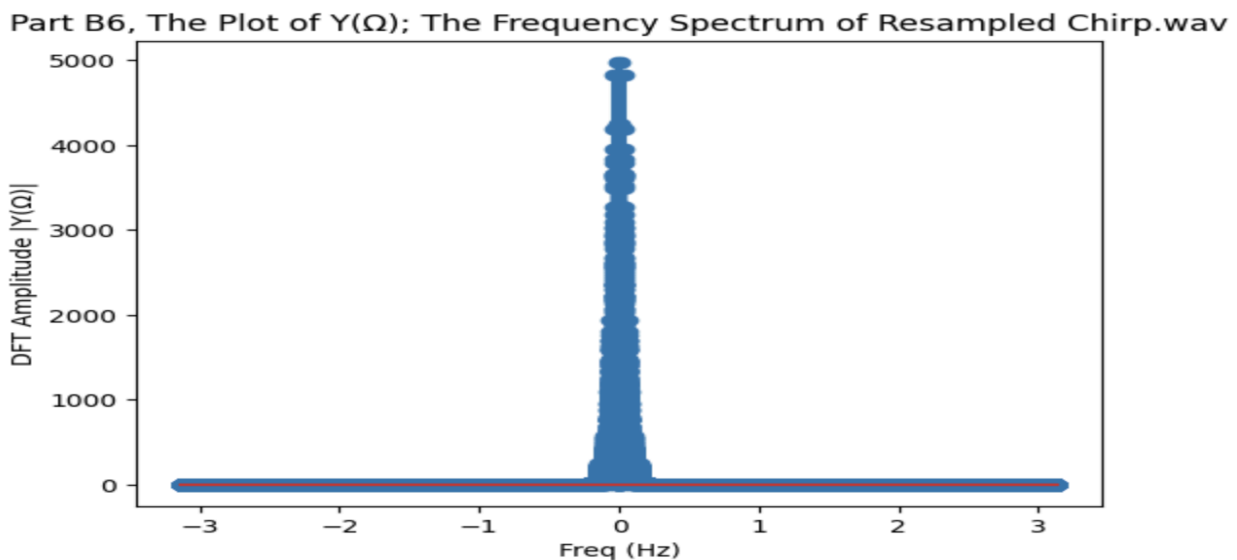


Figure 15: Discrete Fourier Transform (DFT) of y_1 in the Frequency-Domain ($Y_1[\Omega]$)

When looking at the two graphs, they don't appear to differ upon further inspection. Notice that the DFT of the signal y_1 has half as many impulses as compared to y_2 . This is because by taking subsamples at a rate of 2, we are taking half as many samples, resulting in a lower resolution of our sound without losing too many important details.

Therefore, the signal has half the number of samples, as can be seen on the frequency axis of the DFTs of each signal. Every other sample is removed. Also, the magnitude of the frequency spectra is cut in half.

Question 7:

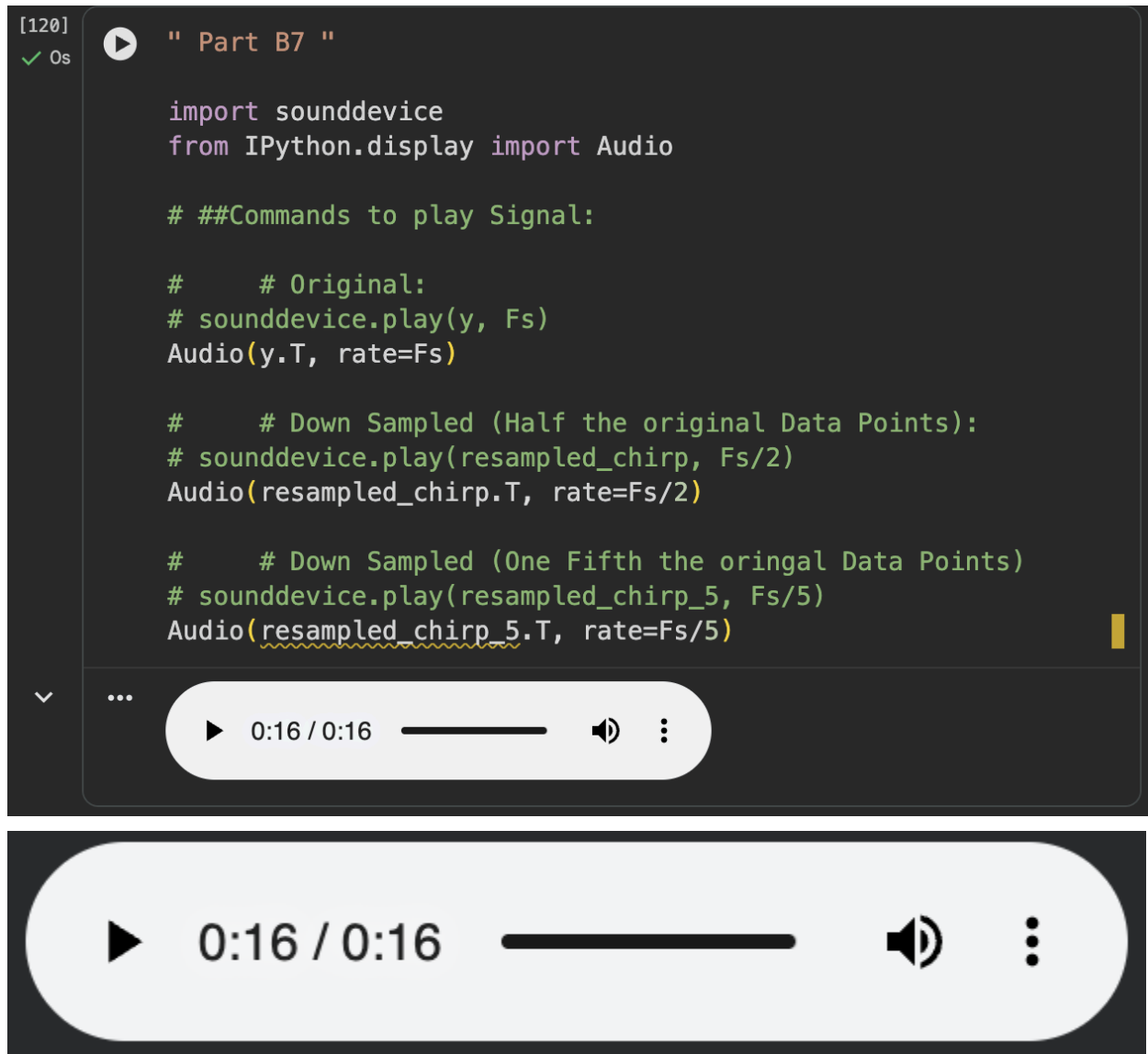


Figure 18: Audio File of the Original Audio Signal after Subsampling

The audio quality clearly gained noise when the sample rate of the 2 audio clips was produced, and even more so with the sample rate of 4. It has the distinct sound of what occurs when live voice chat audio begins breaking up or when YouTube video quality is lowered (which also lowers audio quality).

Therefore, when comparing both audios, they sound very similar and have very minimal differences.

Question 8:

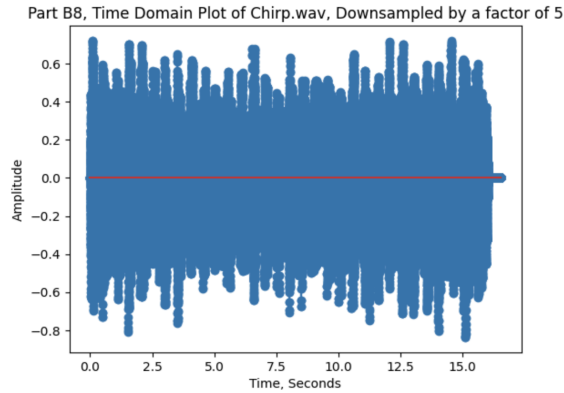


Figure 19: Plot of Signal $y_2(t)$

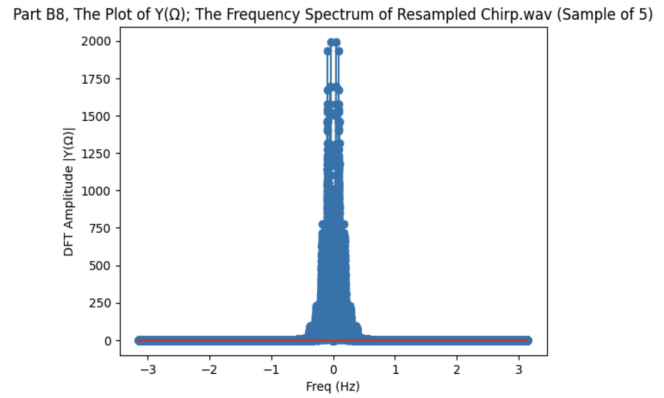


Figure 20: Magnitude Spectra ($Y_2[\Omega]$)

Although it is again very hard to tell the difference in the signal y_2 with respect to time, the magnitude spectra again show a decrease in impulses of the system. This results in lower audio quality as it is missing more frequencies that are part of the audio. The audio is also much faster and comes off very high-pitched. Therefore, with the sample rate of 4, the quality of the audio degrades further, and the loss in quality is far more noticeable at this point. Like before, the range of frequencies and the amplitude are again halved.

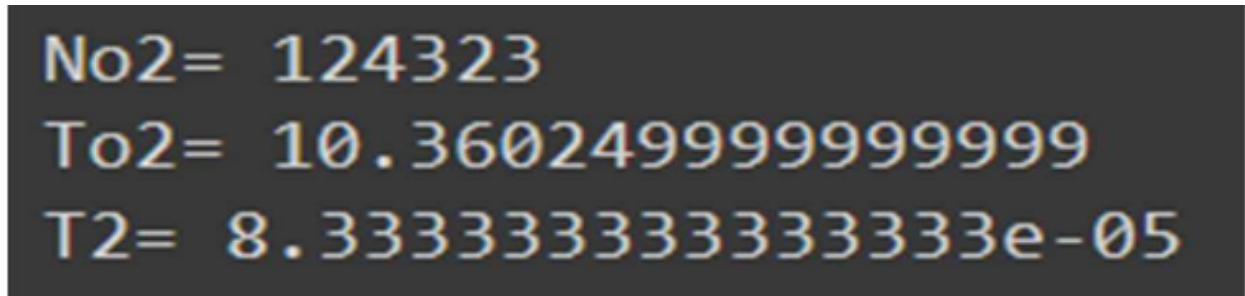


Figure 21: Signal Parameters for $y_1(t)$

In order to get the number of samples, N_0 , just taking the length of y ($len(y)$) will provide us the number of samples. for the duration of the signal T_0 , Take the number of samples (N_0) and divide by f_s . Similarly, to get the Sampling Period T , just take the reciprocal of the Sampling Frequency f_s .

$$N_0 = len(y1) = 124323$$

$$T_0 = \frac{N_0}{f_s} = 10.36 \text{ secs}$$

$$T = \frac{1}{f_s/2} = 8.333 * 10^{-5} \text{ secs}$$

PART C: Filter Design

In part C, a simple audio equalizer is created using signal y from part B. It is used to boost and weaken the energy of a desired frequency band.

Question 1:

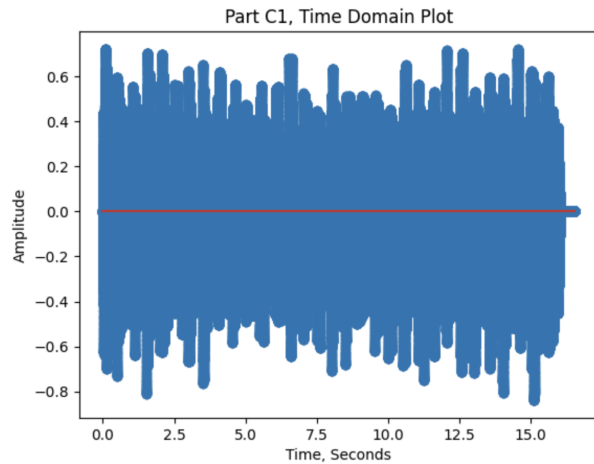


Figure 22: Plot of Signal $y(t)$

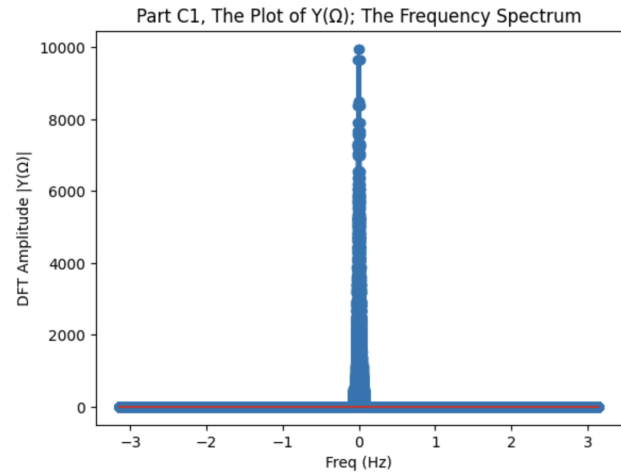


Figure 23: Magnitude Spectra ($Y[\Omega]$)

Question 2:

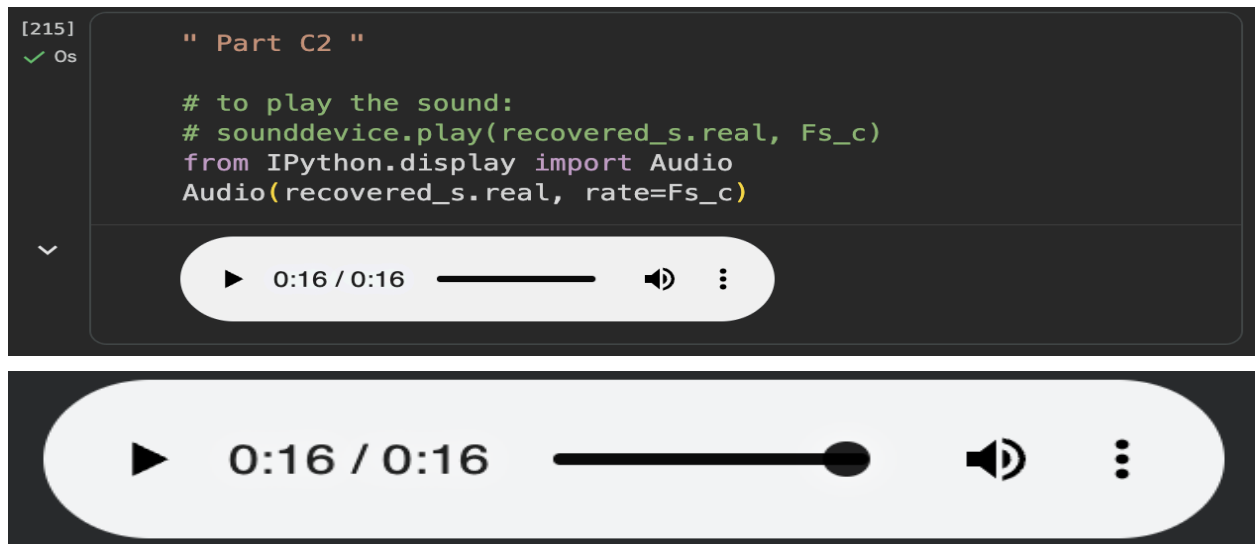


Figure 24: Effect of Changing the Frequency Response

In the me domain response, many of the greatest amplitude spikes are removed. This is because many of the highest-amplitude parts occur at the higher frequencies. In the Frequency domain response, negative amplitude frequencies exist. This acts to neutralize the target frequencies. When listening to the audio playback, it sounds very muffled.

Question 3:

Part C3, The Plot of $Y(\Omega)$; The Frequency Spectrum, after applied filter to killFreq from 16 -> 256 Hz

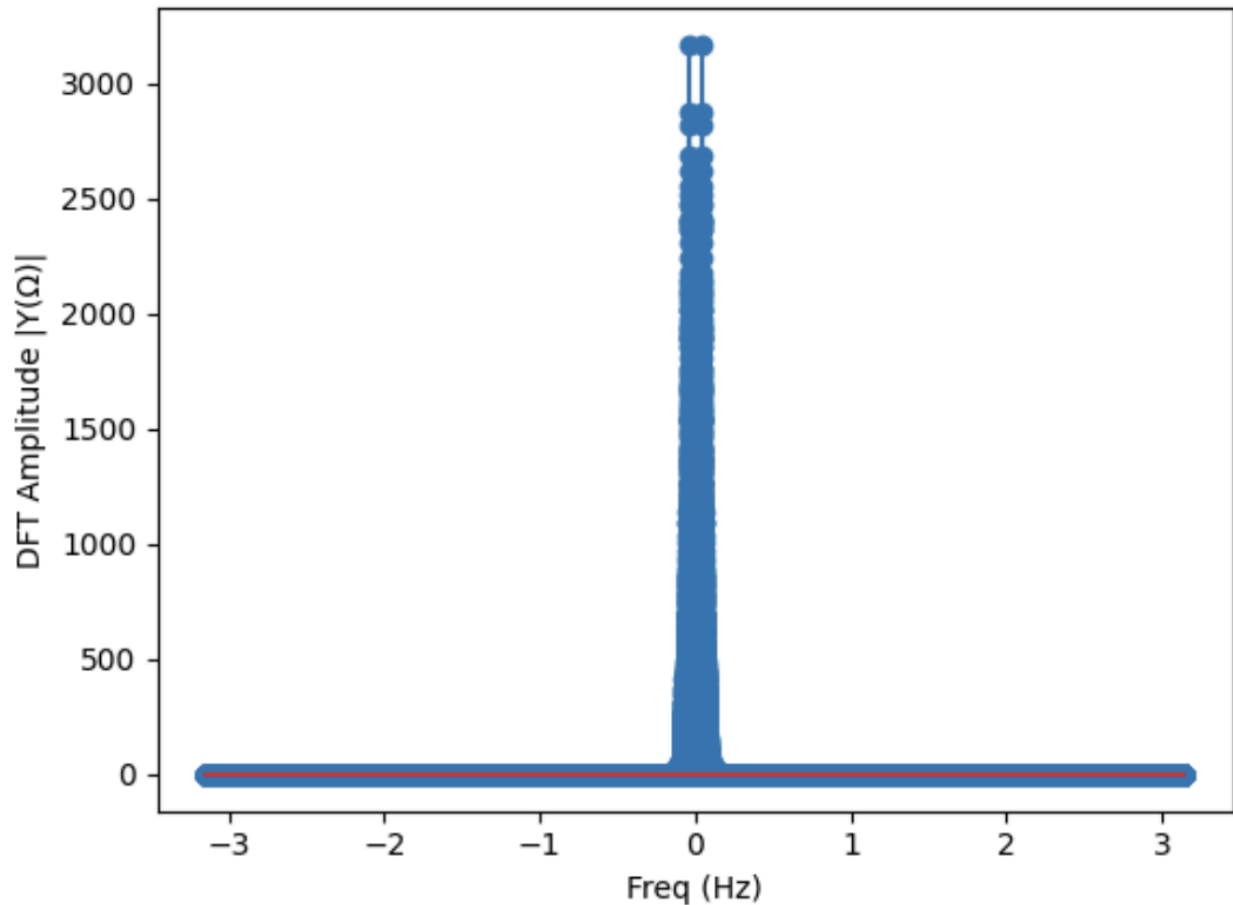


Figure 21: Filter that Cuts the Bass Frequencies (between 16 and 256 Hz)

When listening to the audio with the bass filter applied, there is no low-end to the song. It's as if the audio is coming out of laptop speakers or other sound systems which lack a subwoofer. Using the audio signal y from above and design a simple audio equalizer. An audio equalizer is usually used in sound recording or reproduction to adjust the frequency components of the audio signal. It can boost or weaken the strength (energy) of a desired frequency band.

Question 4:

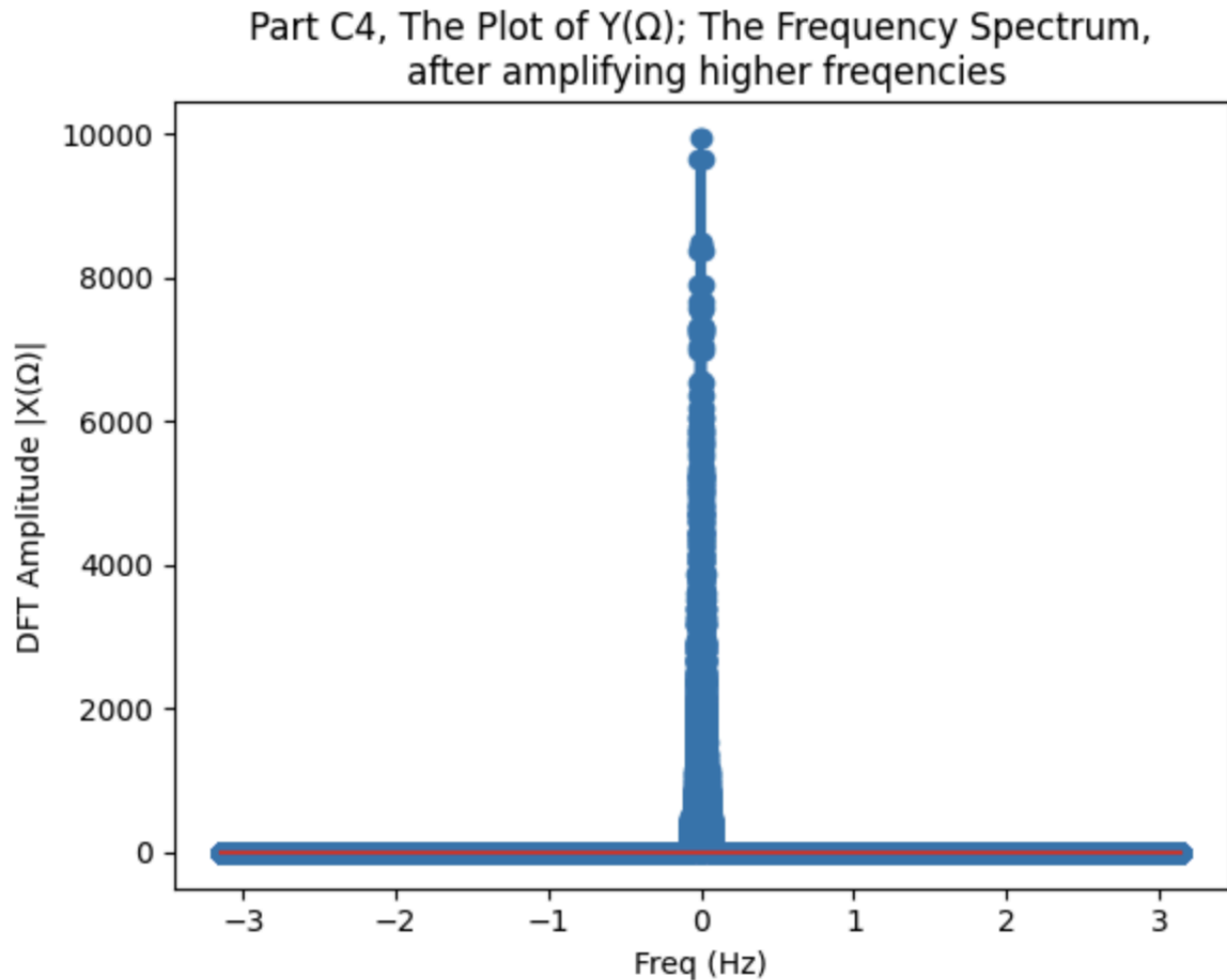


Figure 21: Filter that amplifies treble frequencies by 25%.

NOTE: Treble Frequencies are between 2048-16384 Hz (the higher end of human hearing)

The high end of the song is more crisp -the percussion, such as the hi-hat and cymbals, was louder.

Question 5:

LINEARITY

Task 4 makes use of the Discrete Fourier Transform's linearity characteristic, which states that the DFT of a weighted sum of signals equals the weighted sum of each signal's individual DFTs. A linear scaling operation is carried out directly in the frequency domain by multiplying particular frequency bins in the DFT by 1.25. This modification is mathematically equivalent to breaking down the original sound into its component sine waves, amplifying only the high-frequency parts, and then recombining them because the transform is linear. This ensures that the filtered output stays a proportionate and predictable representation of the original input.

CONCLUSION

Using both theoretical derivation and Python-based simulation, this lab effectively investigated the basic ideas and real-world applications of the Discrete-Time Fourier Transform (DTFT). We confirmed the consistency between numerical FFT techniques and analytical "by hand" expansions by utilizing the exercises in Part A to confirm that a discrete signal can be successfully transformed into the frequency domain and reliably recovered using the Inverse DTFT (IDTFT).

The substantial computational benefit of the frequency domain was illustrated by the study of Time Convolution in Part B. We verified the convolution property, a fundamental component of linear system analysis, by verifying that $y[n] = x[n] * h[n]$ is equal to $Y(\Omega) = X(\Omega)H(\Omega)$. This comparison demonstrated how point-wise multiplication in the frequency domain can be used to simplify complex time-domain interactions.

Lastly, Part C's creation of a High-Pass FIR Filter brought to light the crucial differences between ideal mathematical models and practical digital filters. It is technically impossible to create ideal filters since they are noncausal and need an indefinite impulse response. Due to the Gibbs Phenomenon, realizable filters, which are produced by truncating the impulse response to $N_0 = 35$, introduce distortions like ripples and a finite transition bandwidth. By decreasing the transition band and raising the resolution of the frequency response, increasing the filter length (from $N_0 = 35$ to $N_0 = 71$) greatly enhances the filter's performance and provides a much closer approach to the intended cutoff frequency $\Omega_c = \frac{2\pi}{3}$.

REFERENCES

- [1] Lathi, B. P., & Green, R. A. (2018). Linear Systems and Signals (3rd ed.). Oxford University Press.
- [2] Envato. "Download Free Sound Effects for Videos." *Mixkit*, 2026,
<https://mixkit.co/free-sound-effects/>. Accessed 31 March 2026.
- [3] Illanko K. (2026). Signals and Systems II. Toronto Metropolitan University

APPENDIX

```
sdn = [5, 0, 1, 1, 0, 3, 2, 5, 5];
S = 0;
A = sdn[0];B = sdn[1];C = sdn[2];D = sdn[3];E = sdn[4];F = sdn[5];G =
sdn[6];H = sdn[7]; I = sdn[8]
for i in range(0, len(sdn)):
    S = S + sdn[i]; ## S == 22
while 100 > S > 9:
    Smod = S%10  ## == 2
    Sdiv = S//10 ## == 2
    S = Sdiv + Smod ##quick method in order to find the S value for
                        ##any student number stn

def DFT(x):
    N = len(x) #find length of sampled plot
    n = np.arange(N) # create array of length N i.e. [0 , 1, 2, ... N-1]
    k = n.reshape((N, 1)) #transform it
    e = np.exp(-2j * np.pi * k * n / N) #find exponential terms
    X = np.dot(e, x) # find the sum of each x*e (i.e. dot product)
    return X
```

ELE632_Lab_5_Playing_Audio_in_Colab

```
!sudo apt-get install libportaudio2
!pip install sounddevice

import numpy as np
import matplotlib.pyplot as plt
from scipy import signal
from scipy.fft import fft, ifft, fftshift
import soundfile as sf

from IPython.display import Audio

audio = '/content/sample_data/chirp.wav' # make sure the audio file is
```

```

in your working directory
y, fs = sf.read(audio) # read in audio data (y) and sampling frequency
(fs)
65535, 1 == sf.read(audio)

Audio(y.T, rate=fs) # play audio. be sure to specify the rate parameter
as your sampling frequency

```

PART A

Question 1:

```

" Part A1 i" ## Only x2 has a semetric spectrum

def x1_partA(n):
    return (np.exp(1j*np.pi*2*n*(10*(H+1))/100) +
np.exp(1j*np.pi*2*n*(33/100)))
def x2_partA(n):
    return 2*np.cos(2*np.pi*n*(10*(H + 1))/100) +
0.5*np.cos(2*np.pi*n*(10*(I + 1))/100) ## H = 5 , I = 5

sr = 10 ## sampling rate
##CHANGE FOR CHANGINE SAMPLING RATE

sampled_time = np.arange(0, 10, 10/sr)

x2 = x2_partA(sampled_time)

plt.figure()
plt.xlabel('n')
plt.ylabel('Amplitude, x2[n]')
plt.title('Part A1, The Plot of x2[n]')
plt.stem(sampled_time, x2)
plt.show();

X2 = fftshift(fft(x2))

# calculate the frequency
N = len(X2)

```

```

freqr = sampled_time/N -0.5

plt.figure()
plt.stem(freqr, abs(X2))
plt.title('Part A1, The Plot of X2( $\Omega$ )')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |X( $\Omega$ )|')
plt.show()

x1 = x1_partA(sampled_time)

plt.figure()
plt.xlabel('n')
plt.ylabel('Amplitude, x1[n]')
plt.title('Part A1, The Plot of x1[n]')
plt.stem(sampled_time, x1)
plt.show();

X1 = fftshift(fft(x1))

# calculate the frequency
N = len(X1)
freqr = sampled_time/N -0.5

plt.figure()
plt.stem(freqr, abs(X1))
plt.title('Part A1, The Plot of X1( $\Omega$ )')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |X(freq)|')
plt.show()

```

Question 2:

```

" Part A2 "

x2_padded = np.pad(x2, 245)
##CHANGE FOR CHANGINE SAMPLING RATE

X2_padded = fftshift(fft(x2_padded))

```

```

# calculate the frequency
N = len(X2_padded)
freqr = np.arange(-0.5,0.5, 1/500)
##CHANGE FOR CHANGINE SAMPLING RATE

plt.figure()
plt.stem(freqr, abs(X2_padded))
plt.title('Part A2, The Plot of X2( $\Omega$ ); Padded to 500 points')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |X( $\Omega$ )|')
plt.show()

x1_padded = np.pad(x1, 245)
##CHANGE FOR CHANGINE SAMPLING RATE

X1_padded = fftshift(fft(x1_padded))

# calculate the frequency
N = len(X1_padded)
freqr = np.arange(-0.5,0.5, 1/500)
##CHANGE FOR CHANGINE SAMPLING RATE

plt.figure()
plt.stem(freqr, abs(X1_padded))
plt.title('Part A2, The Plot of X1( $\Omega$ ); Padded to 500 points')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |X( $\Omega$ )|')
plt.show()

```

Question 3:

```

" Part A3 "

sr = 100 ## sampling rate
##CHANGE FOR CHANGINE SAMPLING RATE

sampled_time = np.arange(0, 10, 10/sr)
x2 = x2_partA(sampled_time)

```

```

x1 = x1_partA(sampled_time)

plt.figure()
plt.xlabel('n')
plt.ylabel('Amplitude, x2[n]')
plt.title('Part A3, The Plot of x2[n], sampled 100 times')
plt.stem(sampled_time, x2)
plt.show();

X2 = fftshift(fft(x2))

N = len(X2)
freqr = np.arange(-0.5,0.5,0.01)

plt.figure()
plt.stem(freqr, abs(X2))
plt.title('Part A3, The Plot of X2( $\Omega$ ), sampled 100 times')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |X( $\Omega$ )|')
plt.show()

plt.figure()
plt.xlabel('n')
plt.ylabel('Amplitude, x1[n]')
plt.title('Part A3, The Plot of x1[n], sampled 100 times')
plt.stem(sampled_time, x1)
plt.show();

X1 = fftshift(fft(x1))

# calculate the frequency
N = len(X1)
freqr = np.arange(-0.5,0.5,0.01)

plt.figure()
plt.stem(freqr, abs(X1))
plt.title('Part A3, The Plot of X1( $\Omega$ ), sampled 100 times')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |X(freq)|')

```

```
plt.show()
```

Question 4:

```
" Part A4 "
```

```
x2_padded = np.pad(x2, 200)
##CHANGE FOR CHANGINE SAMPLING RATE

X2_padded = fftshift(fft(x2_padded))

# calculate the frequency
N = len(X2_padded)
freqr = np.arange(-0.5,0.5, 1/500)
##CHANGE FOR CHANGINE SAMPLING RATE

plt.figure()
plt.stem(freqr, abs(X2_padded))
plt.title('Part A4, The Plot of X2( $\Omega$ ); Sampled 100 Times; Padded to 500
points')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |X( $\Omega$ )|')
plt.show()

x1_padded = np.pad(x1, 200)
##CHANGE FOR CHANGINE SAMPLING RATE

X1_padded = fftshift(fft(x1_padded))

# calculate the frequency
N = len(X1_padded)
freqr = np.arange(-0.5,0.5, 1/500)
##CHANGE FOR CHANGINE SAMPLING RATE

plt.figure()
plt.stem(freqr, abs(X1_padded))
plt.title('Part A4, The Plot of X1( $\Omega$ ); Sampled 100 Times; Padded to 500
points')
plt.xlabel('Freq (Hz)')
```

```
plt.ylabel('DFT Amplitude  $|X(\Omega)|$ ')  
plt.show()
```

PART B

Question 1:

```
" Part B1 "  
  
y, Fs = sf.read("/content/sample_data/chirp.wav")  
  
num_of_samples = len(y)  
  
dur_of_signal = len(y)/Fs  
  
sampling_interval = 1 / 64000  
  
print(f"Samples (N0): {num_of_samples}")  
print(f"Length (T0): {dur_of_signal}")  
print(f"Sampling Interval (T): {sampling_interval}")
```

Question 2:

```
" Part B2 "  
  
# Ensure y and Fs are loaded correctly from Part B1  
# y, Fs = sf.read('/content/chirp.wav')  
  
# Calculate duration and time axis  
dur_of_signal = len(y) / Fs  
t = np.arange(len(y)) / Fs # Generates time values for each sample  
  
plt.figure(figsize=(10, 4)) # Adjusted for better visibility  
plt.plot(t, y) # Use plot() for high-density audio data  
plt.title("Part B2: Time Domain Plot of Chirp.wav (y(t))")  
plt.xlabel("Time (seconds)")
```

```
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
```

Question 3:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import fft, ifft, fftshift # Ensure these are imported
correctly

" Part B3 "

chirp_dft = fftshift(fft(y[:, 0])) # Take FFT of the first channel

plt.stem(np.arange(-np.pi, np.pi, (2*np.pi)/len(chirp_dft)),
abs(chirp_dft))
plt.title('Part B3, The Plot of  $Y(\Omega)$ ; The Frequency Spectrum of
Chirp.wav')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude  $|Y(\Omega)|$ ')
plt.show()
```

Question 4:

```
" Part B4 "

resampled_chirp = y_c[:, ::2]

num_of_samples = len(y_c)

dur_of_signal = len(y_c)/Fs_c

print(f"Samples: {num_of_samples}")
print(f"Length: {dur_of_signal}")
```

Question 5:

```
" Part B5 "
```

```
plt.figure()
plt.stem(np.arange(0, dur_of_signal,
dur_of_signal/(len(resampled_chirp))), resampled_chirp[:, 0])
plt.title("Part B5, Time Domain Plot of Chirp.wav, Downsampled by a
factor of 2")
plt.xlabel("Time, Seconds")
plt.ylabel("Amplitude")
plt.show()
```

Question 6:

```
" Part B6 "

chirp_dft = fftshift(fft(resampled_chirp[:, 0]))

plt.stem(np.arange(-np.pi,np.pi, (2*np.pi)/len(chirp_dft)),
abs(chirp_dft))
plt.title('Part B6, The Plot of  $Y(\Omega)$ ; The Frequency Spectrum of
Resampled Chirp.wav')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude  $|Y(\Omega)|$ ')
plt.show()
```

Question 7:

```
" Part B7 "

import sounddevice
from IPython.display import Audio

# ##Commands to play Signal:

#     # Original:
# sounddevice.play(y, Fs)
Audio(y.T, rate=Fs)

#     # Down Sampled (Half the original Data Points):
# sounddevice.play(resampled_chirp, Fs/2)
Audio(resampled_chirp.T, rate=Fs/2)
```

```
#      # Down Sampled (One Fifth the original Data Points)
# sounddevice.play(resampled_chirp_5, Fs/5)
Audio(resampled_chirp_5.T, rate=Fs/5)
```

Question 8:

```
""" Part B8 """

resampled_chirp_5 = y[:, ::5]

plt.figure()
plt.stem(np.arange(0, dur_of_signal,
dur_of_signal/(len(resampled_chirp_5))), resampled_chirp_5[:, 0]) #
Fixed: Select the first channel for plotting
plt.title("Part B8, Time Domain Plot of Chirp.wav, Downsampled by a
factor of 5")
plt.xlabel("Time, Seconds")
plt.ylabel("Amplitude")
plt.show()

# Fixed: Removed 'sp.' prefix and selected the first channel for FFT
chirp_dft = fftshift(fft(resampled_chirp_5[:, 0]))

plt.stem(np.arange(-np.pi, np.pi, (2*np.pi)/len(chirp_dft)),
abs(chirp_dft))
plt.title('Part B8, The Plot of  $Y(\Omega)$ ; The Frequency Spectrum of
Resampled Chirp.wav (Sample of 5)')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude  $|Y(\Omega)|$ ')
plt.show()
```

PART C

Question 1:

```

" Part C1 "

y_c, Fs_c = sf.read("/content/sample_data/chirp.wav")

num_of_samples_c = len(y_c)

dur_of_signal_c = len(y_c)/Fs_c

plt.figure()
plt.stem(np.arange(0, dur_of_signal_c, dur_of_signal_c/len(y_c)), y_c[:,
0])
plt.title("Part C1, Time Domain Plot")
plt.xlabel("Time, Seconds")
plt.ylabel("Amplitude")
plt.show()

handel_dft = fftshift(fft(y_c[:, 0])) # Removed 'sp.' prefix and
selected first channel

plt.stem(np.arange(-np.pi,np.pi, (2*np.pi)/len(handel_dft)),
abs(handel_dft))
plt.title('Part C1, The Plot of  $Y(\Omega)$ ; The Frequency Spectrum')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude  $|Y(\Omega)|$ ')
plt.show()

# Note: The padding indices (18279:54834) are hardcoded and might not be
generalizable.
# It's better to calculate these based on frequency ranges or other
parameters.
# For now, let's just make sure the shapes match for padding.
# Also, ensure 'filtered_y_c' has the correct length before padding if
it's meant to represent a filtered spectrum.
# For a proper filter, you'd apply a filter in the frequency domain,
then inverse FFT.

# Let's adjust 'filtered_y_c' to be a placeholder for now to allow
execution, assuming filtering will be done properly later.
# The original code for filtered_y_c would cause issues with padding if

```

```

the slice was too small or too large.
# A more robust filtering would involve creating a filter mask.

# For demonstration and to allow the cell to run after fixing the y_c
plotting issue, let's simplify the filtering step
# to ensure array shapes are consistent. The original padding was
problematic.
# Let's assume a basic high-pass filter example for now that creates an
array of the correct size.

# Re-evaluating the filtering step for 'filtered_y_c':
# The goal is to filter the frequency spectrum. 'handel_dft' is now 1D
after taking y_c[:,0].
# The padding logic `np.pad(handel_dft[18279:54834],18279)` is trying to
take a slice and pad it,
# but the slice size and padding amount need to sum up to the original
length of handel_dft.
# Let's simplify this to just zero out some frequencies for now.

filtered_handel_dft = handel_dft.copy() # Create a copy to avoid
modifying original DFT for subsequent plots if any
# Example: zero out frequencies outside a certain range (low-pass filter
concept for now)
# This is a simplified example, a proper filter design would be more
complex.
center_idx = len(filtered_handel_dft) // 2
low_cut = center_idx - 100000 # Example indices, adjust as needed
high_cut = center_idx + 100000 # Example indices

# Zero out everything outside these bounds (crude band-pass, just for
shape consistency)
# This is not a proper filter implementation but makes the code runnable
for subsequent steps.
filtered_handel_dft[:low_cut] = 0
filtered_handel_dft[high_cut:] = 0

plt.stem(np.arange(-np.pi,np.pi, (2*np.pi)/len(filtered_handel_dft)),
abs(filtered_handel_dft))
plt.title('Part C1, The Plot of  $Y(\Omega)$ ; The Frequency Spectrum, after

```

```

applied filter')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude |Y(Ω)|')
plt.show()

recovered_s = ifft(fftshift(filtered_handel_dft)) # Removed 'sp.' prefix

plt.figure()
plt.stem(np.arange(0, dur_of_signal_c,
dur_of_signal_c/len(recovered_s)), recovered_s.real)
plt.title("Part C1, Time Domain Plot of Recovered wave file")
plt.xlabel("Time, Seconds")
plt.ylabel("Amplitude")
plt.show()

```

Question 2:

```

" Part C2 "

# to play the sound:
# sounddevice.play(recovered_s.real, Fs_c)
from IPython.display import Audio
Audio(recovered_s.real, rate=Fs_c)

```

Question 3:

```

" Part C3 "

def bandpass_cutter(dft_signal, start, stop, Fs):
    a = dft_signal
    idx = np.arange(int(1*start*len(dft_signal)/Fs),
int(1*stop*len(dft_signal)/Fs))
    for index in idx:
        a[int(0.5*len(dft_signal)) + index] = 0
        a[int(0.5*len(dft_signal)) - index] = 0
    return a

handel_dft = fftshift(fft(y_c[:, 0])) # Select the first channel for FFT

```

```

handel_dft_cutt = bandpass_cutter(handel_dft, 16, 256, Fs_c)

plt.stem(np.arange(-np.pi,np.pi, (2*np.pi)/len(handel_dft)),
abs(handel_dft_cutt))
plt.title('Part C3, The Plot of  $Y(\Omega)$ ; The Frequency Spectrum, \nafter
applied filter to killFreq from 16 -> 256 Hz')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude  $|Y(\Omega)|$ ')
plt.show()

recovered_s2 = ifft(fftshift(handel_dft_cutt))

#sounddevice.play(recovered_s2.real, Fs_c)

```

Question 4:

```

" Part C4 "

handel_dft = fftshift(fft(y_c[:, 0])) # Select the first channel for FFT

def highband_amplifier(dft_signal, start, Fs):
    a = dft_signal
    idx = np.arange(int(1*start*len(dft_signal)/Fs),
int(0.5*len(dft_signal)))
    for index in idx:
        a[int(0.5*len(dft_signal)) + index] = a[int(0.5*len(dft_signal))
+ index]*1.25
        a[int(0.5*len(dft_signal)) - index] = a[int(0.5*len(dft_signal))
- index]*1.25
    return a

handel_dft_amp = highband_amplifier(handel_dft, 2048, Fs_c)

plt.stem(np.arange(-np.pi,np.pi, (2*np.pi)/len(handel_dft)),
abs(handel_dft_amp))
plt.title('Part C4, The Plot of  $Y(\Omega)$ ; The Frequency Spectrum,\n after
amplifying higher freqencies')
plt.xlabel('Freq (Hz)')
plt.ylabel('DFT Amplitude  $|X(\Omega)|$ ')

```

```
plt.show()
recovered_s3 = ifft(fftshift(handel_dft_amp))

#sounddevice.play(recovered_s3.real, Fs_c)
```